

Amazon Customer Review Video Game Analysis

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ABSTRACT

This project uses network analysis to study customer reviews of Amazon video games from 1995 to 2013, taken from the SNAP dataset. The reviews are modelled as a bipartite network of customers and games. For each star rating, the user and product projections are compared at two thresholds, min3 and min5. The results show a clear pattern. Higher ratings produce more connected networks, while the 1 and 2-star networks stay sparse and almost empty. The networks get steadily more connected up to the 5-star rating. A few popular games act as hubs that link different groups of players. The threshold also matters: the stricter min5 rule removes the weaker links and leaves a small, tightly connected core. Overall, the project shows that looking at reviews as a network can reveal patterns that are hard to see when reviews are read one by one.

KEYWORDS

Complex Networks, Bipartite Graph, Projection Sets, Customer Star Rating, Thresholding, Degree, Betweenness Centrality, Closeness Centrality, Harmonic Closeness Centrality, Page Rank.

1. INTRODUCTION

This project aims to study customer reviews on Video Games listed on Amazon from 1995 - 2013. The dataset used for this project was obtained from Stanford Large Network Dataset Collection (SNAP). The analysis maps relationships between Video Game Titles and Customer Profile Names.

Objective

This project aims to conduct a comparative analysis within each Star Rating for Profile Names and Video Game Titles (min3 vs min5 shared neighbours) using Structural Properties, Node-level Metrics, Global Metrics, Community Detection, Largest Connected Component Metrics and Centralities. The project will also cover comparison of the different Star Ratings.

Bipartite Graph

This analysis of this project is conducted on the projections of a Bipartite Graph with edges between Profile Names and Video Game Titles. The nodes within each set of the Bipartite Graph don't share edges among themselves and this is what makes them Bipartite, i.e a Profile Name can only review a Video Game Title and not another Profile Name. The projection sets created from this Bipartite graph are the U projection : User Profile – User Profile and the V Projection : Video Game Titles – Video Game Title,

where the criteria for nodes within each of these projection sets to share an edge is that they should share at least 3 and 5 common neighbours in the corresponding set.

Dataset Statistics

The original dataset contained 463,669 reviews in a text file and this dataset was cleaned to remove unknown/null values from either the User Profile Name or Video Game Titles. The cleaned dataset contained 363,866 reviews.

2. RELATED WORK

Prior research relevant to this project falls into few broad areas: studies of online review helpfulness, network-based approaches to recommendation, the foundations of complex-network analysis, and network analysis applied directly to online reviews. This section reviews each area in turn and identifies the gap this project addresses.

Mudambi and Schuff [1] studied what makes an online review helpful to customers. They found that helpfulness depends on how extreme the rating is, how detailed the review is, and what type of product is being reviewed. This was further explored further by studies [2, 3] that focused on review text and content. Bilal et al. [4] went further by also looking at how the

reviewer's position in a social network affects helpfulness. However, most of these research studies reviews individually based on the features of each review (length, text, its rating) rather than looking at how reviews or reviewers connect to each other in a network.

McAuley and Leskovec [5] used online reviews to recommend products to users. Their focus was on predicting what a person would enjoy based on their past reviews and how their taste changes over time. Jamali and Ester [6, 7] also worked with networks, using trust between users to recommend items. These studies show that reviews can be used as network data, but mainly to make recommendations rather than to study the structure of the network.

Albert and Barabási [8] showed that many real networks are "scale-free," meaning a few nodes have far more connections than the rest. These highly connected nodes are called hubs. Newman [9] and Boccaletti et al. [10] give a broad overview of complex networks and explain the main measures used to study them, such as degree, clustering, path length, centrality, and communities. This project uses these same measures to analyse the review network.

Wang [11] used network analysis on online reviews to divide customers into market segments. This work is the closest to this project because it also treats reviews as a network instead of just text. However, Wang built a single network and did not compare different rating levels or connection thresholds. Other studies also analysed online reviews but for different purposes. Xu et al. [12] mined review text for business insights, and Zhang et al. [13] surveyed how the field of review research has developed over time. None of these studies compare how the network changes across different star ratings and connection thresholds. This project fills that gap by building bipartite projections for each star rating and comparing them at two thresholds (min3 and min5) to see how the structure and the hubs change.

3. METHODOLOGY

Pre-processing

The initial raw dataset contained 463,669 reviews in a text file which was converted into a CSV format file with 5 columns – Product ID, Product Title, UserID, Profile Name and Review Score using Pandas DataFrame for further processing. The records in this CSV format file were cleaned by removing Null/Unknown values, resulting in a CSV format file with 363,866 rows. 5 different Excel sheets were

created from this CSV format file containing the data for each Star Rating.

Projection Sets

The 5 Excel sheets were used to build a Base Bipartite Edge table using DuckDB, where data was stored directly in RAM. This base Bipartite Edge table ensured there were no duplicate reviews from a Profile Name to a Video Game Title using DISTINCT function. The conditions given for a projection set to share an edge were at least 3 & 5 common neighbours – done by SQL statement using GROUP BY and HAVING COUNT (*) >= n. The resultant output was 4 CSV-format files: projection U – Edge and Node table files, and projection V – Edge and Node table files. This process was repeated for all 5 Star Rating categories individually. For computation in Network and graph visualization in Gephi, the projection Edge and Node table files were updated in Excel using lookups.

Metric Computations

The script accepts all projection Edge and Node tables, then ensures every node file has a matching edge file and vice versa, based on the file Naming Convention. Python datatypes - lists, sets and tuples were used to accomplish this. Closeness and Betweenness centrality were calculated on the Largest Connected Component to avoid long runtimes. Also, limiting these centrality measures to LCC guarantees that every node can actually reach every other, making these scores meaningful. Using Networkx built-in functions – all node-level Centrality Metrics, Average Density, Average Clustering Coefficient, Global Efficiency, Average shortest path length, Diameter and PageRank were computed. Top_node was used to find the node with highest value in each metric column. Closeness and Betweenness centrality were calculated on the Largest Connected Component to avoid long runtimes. Networkx built-in functions were used to calculate these metrics.

Result Validation

There were 3 metrics used to validate the output results computed by NetworkX -

1. Degree Sum = 2 * No of Edges.
2. Sum of Connected Components = Total number of Nodes.
3. Average Clustering and Global Efficiency Values must be between 0 and 1.

4. RESULTS

1-2 Star Network Min 3 vs Min 5 Neighbour's

With thresholds of min 3 and 5, the networks for 1 & 2 star rating are sparse and small. The density and clustering is near zero for both U and V projections. here exists no path between the nodes, forming no communities between them, and LCC paths are only 1-2 hops long.

3 Star Network Min 3 vs Min 5 Neighbor's

With thresholds of min 3 and 5 networks, 3 Star Rating graphs are still sparse but better than 1 & 2 Star. Clustering and Global efficiency are still extremely low. The Average Path Length is low, implying that the graph is sparse.

4 Star Network Min 3 vs Min 5 Neighbor's

With thresholds of min 3 and 5, networks for 4 Star Rating are larger. The projection V(Video Game Title-
NetworkX Results

5 Star

Vide Game Title) has a higher density showing tighter clusters compared to U(Profile Name to Profile Name).Now 4 star have more stable, well-connected communities shown by higher global efficiency and average shortest path length scores.

5 Star Network Min 3 vs Min 5 Neighbor's

High global efficiency indicates that these clustered groups remain well connected across the full graph. It has the strongest local grouping, as reflected by its high average clustering. In 5 star, we can identify high-volume reviewers like 'Lisa Shea' having extremely high degree and page rank. 'Lisa Shea' also acts as a hub that connects many separate clusters. Products like 'GTA III', Halo: Combat Evolved' are the most popular nodes with high degree, PageRank and betweenness in projection set V.

Table 1: NetworkX 5 star

Rating-Proj	Threshold	Density	Avg clustering	Global efficiency	LCC avg path	LCC diam.
5★-u	min3	~1.79e-6	0.00839	~1.08e-4	3.16	8
5★-u	min5	~1.04e-7	0.00088	~2.9e-6	2.70	6
5★-v	min3	~6.02e-5	0.03432	~2.34e-3	3.36	9
5★-v	min5	~1.60e-5	0.00042	~4.2e-4	3.49	9

Table 2 : NetworkX 5 star

Ratin g-Proj	Node s	Edg es	Top degree node	Deg ree	Top harmoni c node	Harmo nic centrali ty	Top closeness LCC node	Closen ess LCC	Top betweenness LCC node	Betweenness LCC
5★-u min 3	130,814	15,341	Lisa Shea	831	Lisa Shea	1543.08	Lisa Shea	0.5753	Lisa Shea	0.5085

5★-u min 5	130,8 14	888	Lisa Shea	190	Lisa Shea	267.67	Lisa Shea	0.6635	Lisa Shea	0.6343
5★-v min 3	15,86 3	7,57 8	Grand Theft Auto III	239	GREATER ST HITS: Grand Theft Auto III	742	GREATER ST HITS: Grand Theft Auto III	0.485	Halo: Combat Evolved	0.1407
5★-v min 5	15,86 3	2,01 8	Halo: Combat Evolved	92	Halo: Combat Evolved	300.85	Halo: Combat Evolved	0.4525	Halo: Combat Evolved	0.2119

4 Star

Table 3 : NetworkX 4 star

Rating-Proj	Threshold	Density	Avg clustering	Global efficiency	LCC avg path	LCC diam.
4★-u	min3	~2.9e-7	0.00085	~7.1e-6	3.97	11
4★-u	min5	~3.1e-8	0.00013	~1.5e-7	3.16	7
4★-v	min3	~6.4e-6	0.00454	~9.1e-5	5.40	12
4★-v	min5	~6.1e-7	0.00075	~8.7e-7	1.93	4

Table 4 : NetworkX 4 star

Rating-Proj	Nodes	Edges	Top degree node	Degree	Top harmonic node	Harmonic centrality	Top closeness LCC node	Closeness LCC	Top betweenness LCC node	Betweenness LCC
4★-u min 3	58,28 8	491	Lisa Shea	55	Lisa Shea	155.15	Lisa Shea	0.45	Lisa Shea	0.4829

4★-u min 5	58,28 8	53	Lisa Shea	15	Lisa Shea	23.62	Lisa Shea	0.5303	Lisa Shea	0.5434
4★-v min 3	13,16 5	551	Super Mario Sunshi ne	23	Kingdo m Hearts	92.15	Kingdo m Hearts	0.2916	Kingdom Hearts	0.3924
4★-v min 5	13,16 5	53	Final Fantas y X	5	Final Fantasy X	6	Final Fantasy X	0.7778	Final Fantasy X	0.7143

3 Star

Table 5 : NetworkX 3 star

Rating-Proj	Threshold	Density	Avg clustering	Global efficiency	LCC avg path	LCC diam.
3★-u	min3	~6.6e-8	~2.1e-5	~1.1e-7	2.24	4
3★-u	min5	~5.1e-9	0	~6.3e-9	1.33	2
3★-v	min3	~6.8e-7	0	~7.9e-7	2.0	4
3★-v	min5	~4.0e-8	0	~4.0e-8	1.0	1

Table 6 : NetworkX 3 star

Rating-Proj	Nodes	Edges	Top degree node	Degree value	Top harmonic node	Harmonic centrality value
3★-u min 3	28,080	26	N. Durham "Big Evil"	5	N. Durham "Big Evil"	7
3★-u min 5	28,080	2	N. Durham "Big Evil"	2	N. Durham "Big Evil"	2
3★-v min 3	9,968	34	Necronomicon	2	Dead or Alive 3	3

3★-v min 5	9,968	2	Xbox LIVE 3 Month Gold Membership	1	Xbox LIVE 3 Month Gold Membership	1
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1 & 2 Star

Table 7 : NetworkX 1&2 star

Rating-Proj	Threshold	Density	Avg clustering	Global efficiency
1★-u	min3	$\sim 3.5e-7$	$\sim 7e-4$	$\sim 5e-7$
1★-u	min5	$\sim 6.5e-9$	$\sim 8.6e-5$	$\sim 7e-9$
1★-v	min3	$\sim 6.8e-7$	$\sim 5.3e-4$	$\sim 1.3e-6$
1★-v	min5	$\sim 4.5e-7$	$\sim 8.9e-1$	$\sim 4.8e-7$
2★-u/v	min3/5	≈ 0	0	0

Table 8: NetworkX 1 star

Rating-Proj	Nodes	Edges	Top degree node	Degree	Top harmonic node	Harmonic centrality
1★-u min3	34,994	213	Steve Smith	28	Steve Smith	28.5
1★-u min5	34,994	4	Michael D. Cooper	2	Michael D. Cooper	2
1★-v min3	9,675	32	Star Wars Galaxies: An Empire Divided	7	Star Wars Galaxies: An Empire Divided	9.67
1★-v min5	9,675	21	Star Wars Galaxies: Starter Kit	5	Star Wars Galaxies: Starter Kit	5

Table 9 : NetworkX 2 star

Rating-Proj	Nodes	Edges
2★-u min 3	19,053	0
2★-u min 5	19,053	0
2★-v min 3	7,558	5
2★-v min 5	7,558	3

Gephi Visualization and Results

Threshold Condition - Minimum 3 neighbours

5 Star

Gephi – Set U

Table 10 : Gephi 5 star Set U min 3

Id	profile name	Degree	Closeness centrality	Harmonic Closeness centrality	Betweenness centrality	Page ranks
130477	Lisa Shea	831	0.575315	0.663406	1375048.226	0.138295
6148	Cloud "..."	355	0.484382	0.542605	273779.4393	0.120469
31	N. Durham "Big Evil"	308	0.487325	0.537281	279729.0665	0.138295
73113	Samario C. Oliver	231	0.448515	0.498567	103096.4847	0.102941
73223	S. Rhodes	221	0.464179	0.508763	140119.9178	0.138295

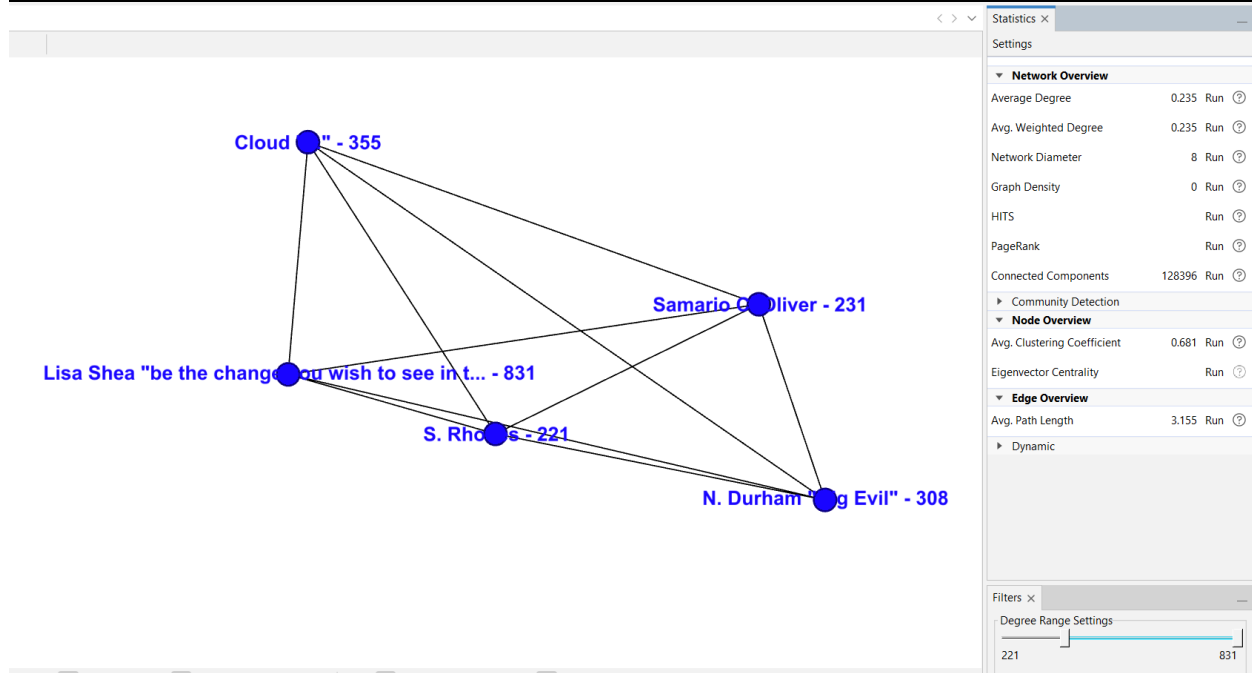


Figure 1 : Gephi 5 Star Set U min 3

Gephi – Set V

Table 11: Gephi 5 Star set V min 3

Id	product title	Degree	Closeness centrality	Harmonic closeness centrality	Betweenness centrality	Page ranks
2151	GREATEST HITS:Grand Theft Auto III	239	0.485023	0.552083	116082.378	0.005221
2103	Halo: Combat Evolved	217	0.466991	0.534226	127006.11	0.005521
14443	Grand Theft Auto Vice City	214	0.475584	0.5391	85574.7894	0.004594
10377	Final Fantasy VII	179	0.461697	0.520548	50167.3616	0.003343
14380	Kingdom Hearts	171	0.450553	0.510503	44722.0186	0.003408
143	Super Smash Bros Melee	164	0.452678	0.509846	55875.3068	0.00331

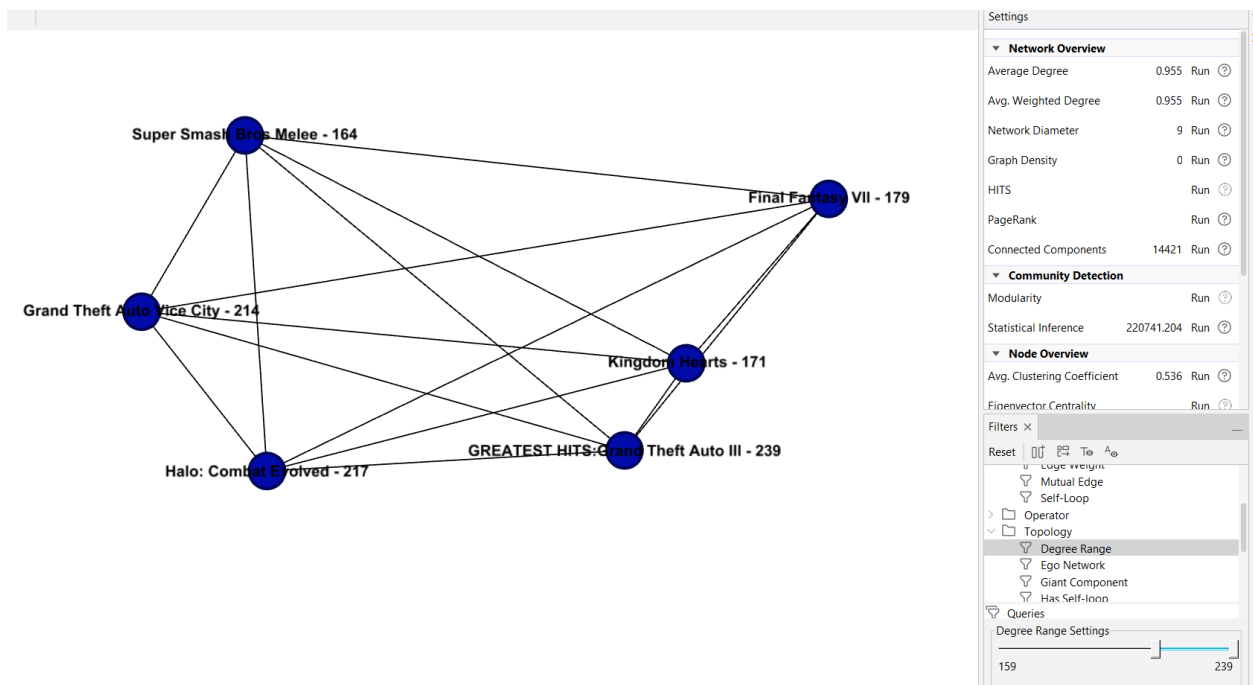


Figure 2 : Gephi 5 Star Set V min 3

4 Star

Gephi – Set U

Table 12 : Gephi 4 star Set U min 3

Id	profile name	Degree	Closeness centrality	Harmonic closeness centrality	Betweenness centrality	Page ranks
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58152	Lisa Shea "	55	0.45	0.538715	19958.886	0.001631
9	Michael Kerner "Michael Kerner"	40	0.372093	0.459962	6389.2011	0.001191
58142	SleepyJD	34	0.379447	0.458168	8776.7969	0.001152
58176	S. Rhodes	29	0.393981	0.469105	5985.0496	0.000804
33	N. Durham "Big Evil"	26	0.390773	0.461756	5098.144	0.000787
52	Trent Foreman	23	0.366412	0.433631	3416.2296	0.000655

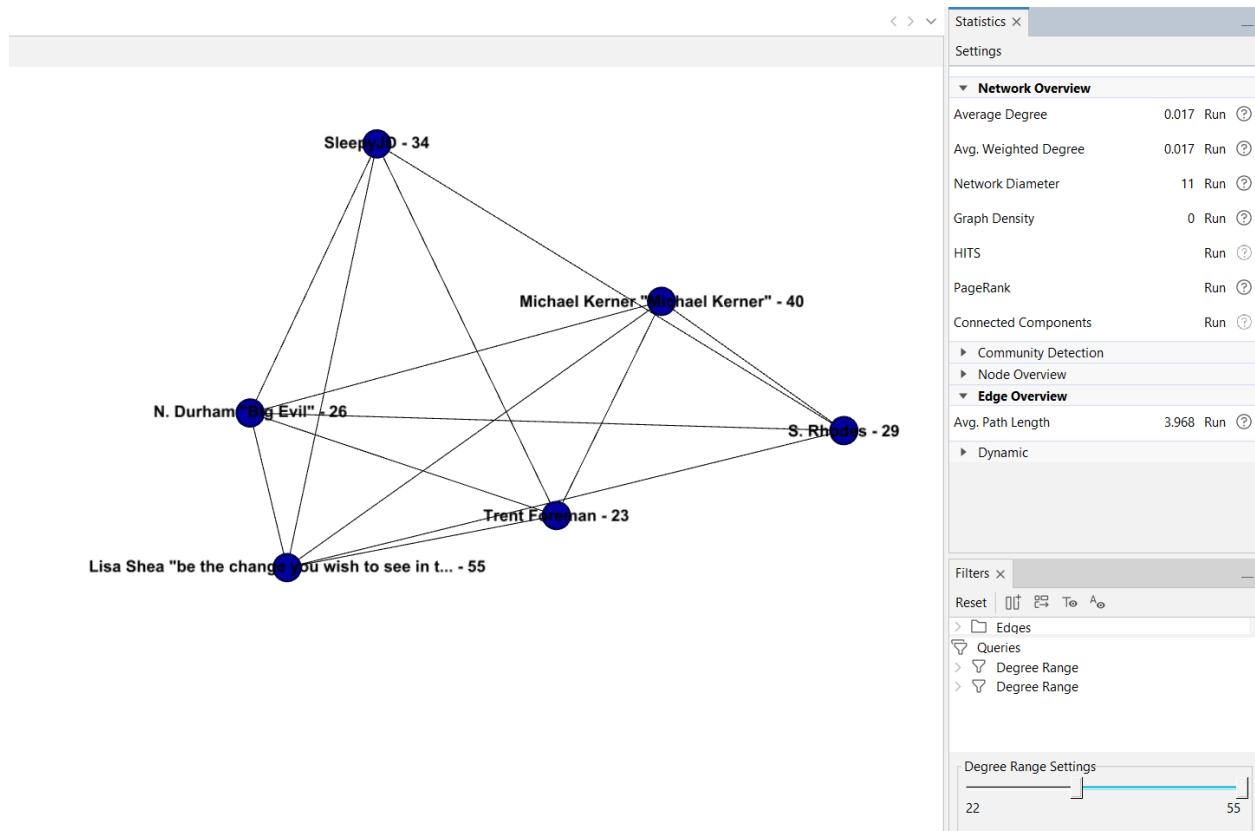


Figure 3 : Gephi 4 Star Set U min 3

Gephi – Set V

Table 13 : Gephi 4 star Set V min 3

Id	product title	Degree	Closeness centrality	Harmonic closeness centrality	Betweenness centrality	Page ranks
6150	Super Mario Sunshine	23	0.257843	0.336221	6122.645346	0.002535
8262	Halo 3	22	0.239308	0.321462	9322.715866	0.002234
8247	Final Fantasy X	18	0.241063	0.304708	3387.942788	0.002263
2075	Kingdom Hearts	15	0.291574	0.350389	13519.69564	0.001779
2086	Big Brain Academy	14	0.212268	0.275629	3252.31816	0.001724
6190	Mario Kart: Double	13	0.249763	0.310601	4175.899857	0.001377

13119	Luigi's Mansion	13	0.229895	0.294908	2227.149744	0.001495
2094	Grand Theft Auto Vice City	12	0.269744	0.31902	6487.984739	0.00158
13114	Condemned Criminal Origins	11	0.204351	0.255758	3951.597259	0.001396

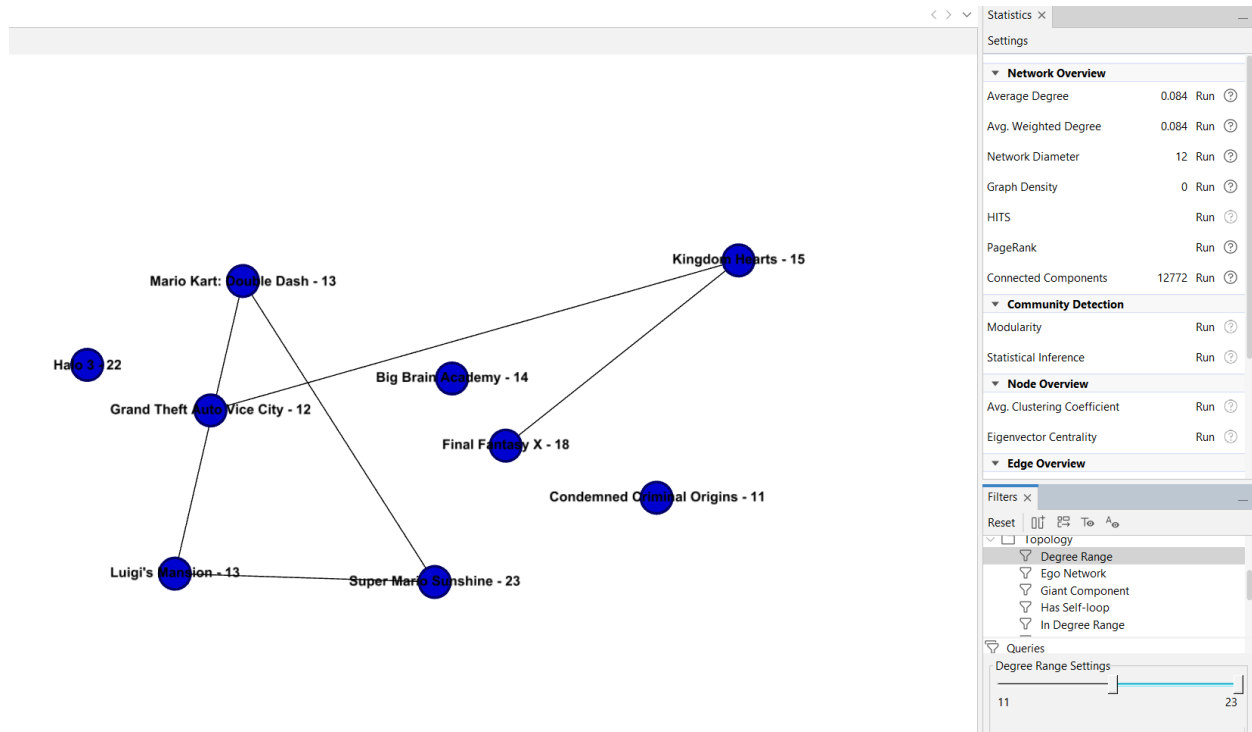


Figure 4 :Gephi 4 Star Set $V \min 3$

3 Star

Gephi – Set U

Table 14 : Gephi 3 star Set $U \min 3$

Id	product title	Degree	Closeness centrality	harmonic closeness centrality	Betweenness centrality	Page ranks
28043	N. Durham "Big Evil"	5	0.692308	0.777778	25	0.000598
28069	Lisa Shea	4	0.6	0.703704	15	0.000483
28073	S. Rhodes	3	0.5625	0.648148	8	0.000363
28056	B. E Jackson	2	0.473684	0.555556	8	0.000271
28051	IRate	2	0.75	0.833333	2	0.000334
28072	wysewomon "wysewomon"	2	0.75	0.833333	2	0.000334
28048	H. A Huffman "haumf"	2	1	1	1	0.000376

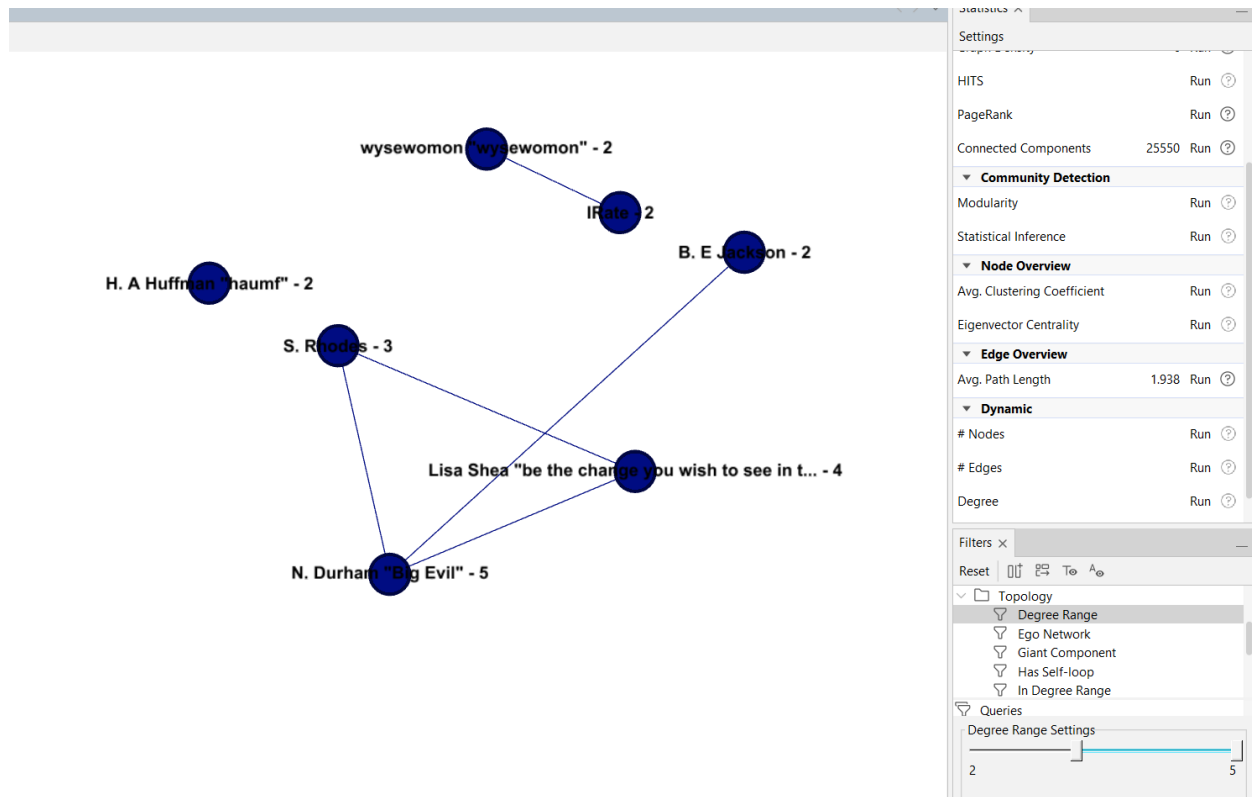


Figure 5 : Gephi 3 Star Set U min 3

Gephi – Set V

Table 15 : Gephi 3 star Set V min 3

Id	Profile Name	Degree	Closeness centrality	Harmonic closeness centrality	Betweenness centrality	Page ranks
9909	Halo 2	2	1	1	1	0.000941
9913	Star Wars: The Force Unleashed	2	1	1	1	0.000941
9918	Dead or Alive Xtreme Beach Volleyball	2	0.571429	0.708333	3	0.000793
9925	Final Fantasy X	2	1	1	1	0.000941
9934	Project Gotham Racing	2	0.571429	0.708333	3	0.000793
9936	Dead or Alive 3	2	0.666667	0.75	4	0.00077
9940	Lost Planet: Extreme Condition	2	1	1	1	0.000941
9951	Necronomicon	2	1	1	1	0.000941
9962	Killzone 2	2	1	1	1	0.000941

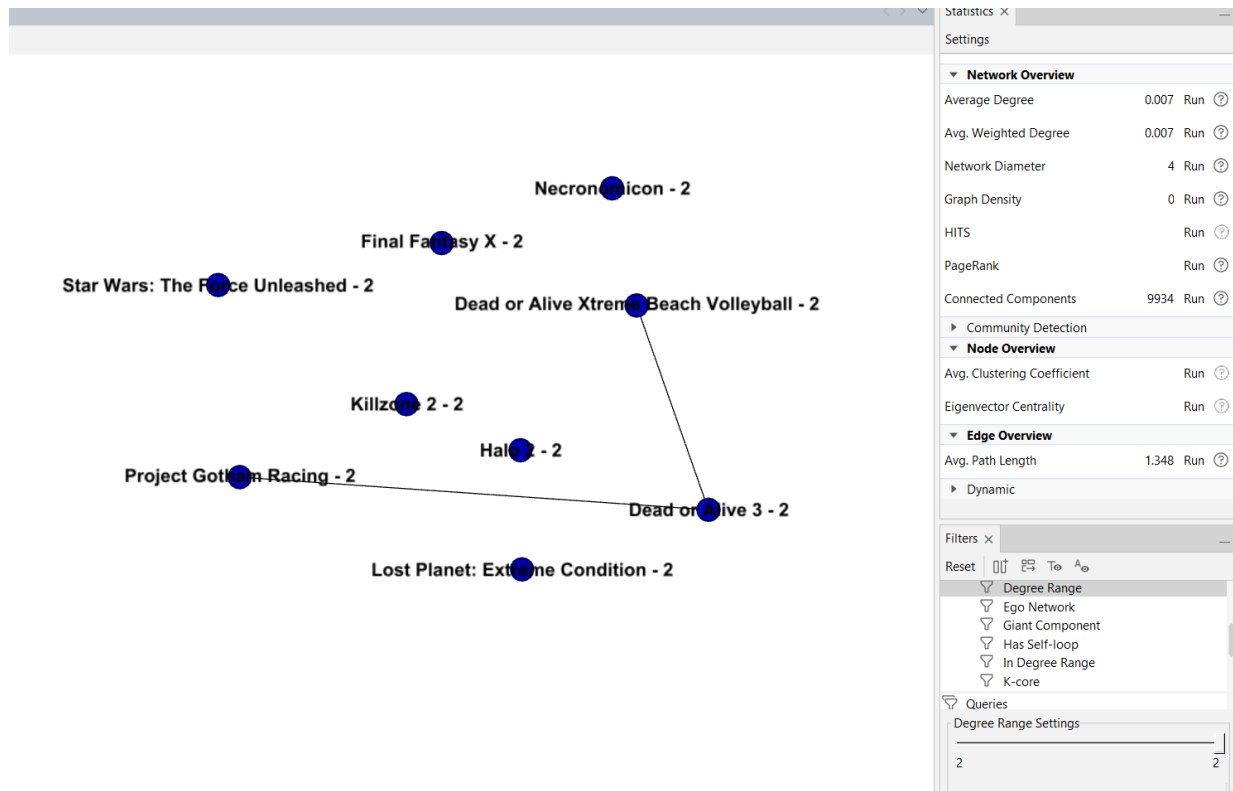


Figure 6 : Gephi 3 Star Set V min 3

2 Star

Set U – no edges

Gephi – Set V

Table 16 : Gephi 2 star Set V min 3

Id	Profile Name	Degree	Closeness centrality	Harmonic closeness centrality	Betweenness centrality	Page ranks
7548	PlayStation 3 (20GB)	1	1	1	0	0.000871
7549	NBA Live 2001	1	1	1	0	0.000871
7550	Dungeons & Dragons Online: StormReach	1	1	1	0	0.000871
7551	Xbox LIVE 3 Month Gold Membership	1	1	1	0	0.000871
7552	Monster Cable PLW100 FO-10 Fiber Optic Audio Hookup for Sony PlayStation 2	1	1	1	0	0.000871

7553	NBA Live 2001	1	1	1	0	0.000871
7554	PlayStation 3 60GB System	1	1	1	0	0.000871
7555	Xbox LIVE 12 Month Gold Membership Card	1	1	1	0	0.000871
7556	Dungeons & Dragons Online: StormReach	1	1	1	0	0.000871
7557	Monster Cable PLW100 FO-10 Fiber Optic Audio Hookup for Sony PlayStation 2	1	1	1	0	0.000871

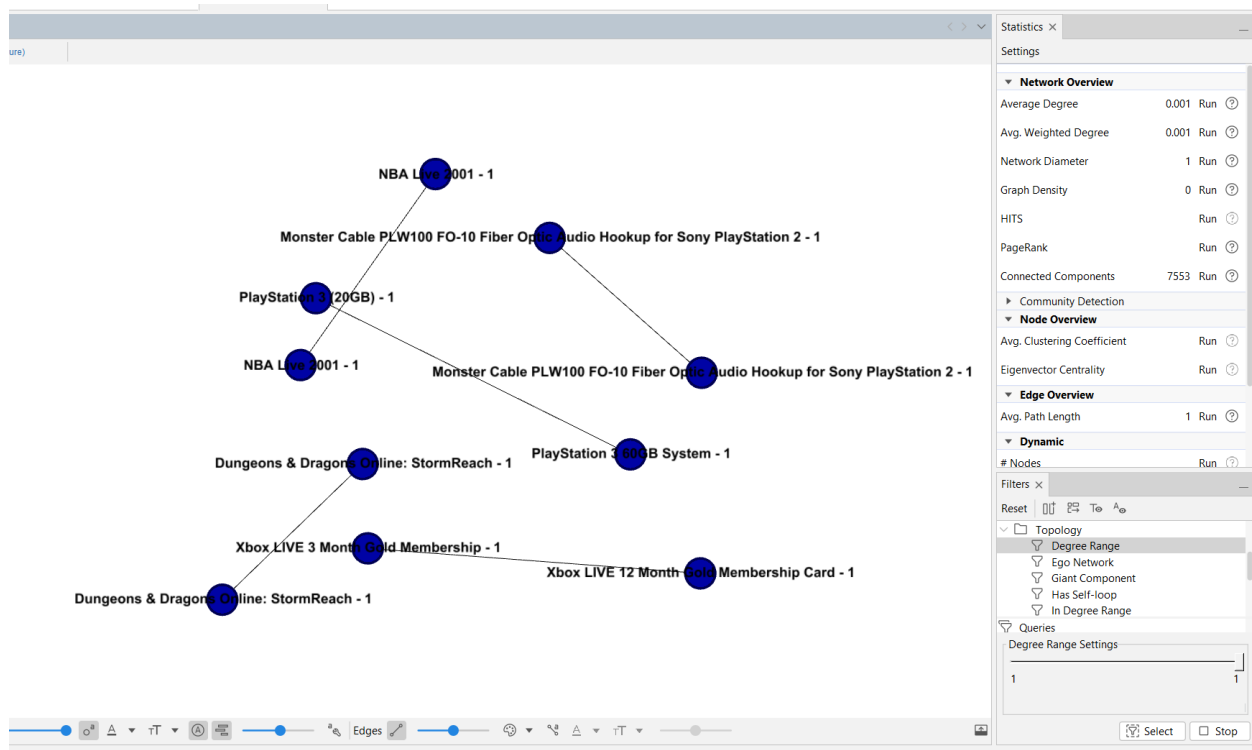


Figure 7 : Gephi 2 Star Set $V_{min} 3$

1 Star

Gephi – Set U

Table 17 : Gephi 1 star Set $U_{min} 3$

Id	profile name	Degree	Closeness centrality	Harmonic Closeness centrality	Betweenness centrality	Page Rank
34971	Steve Smith	28	0.966667	0.982759	45.811905	0.000366
34975	K. Geddings	28	0.966667	0.982759	45.811905	0.000366
34983	Michael D. Cooper	28	0.966667	0.982759	45.811905	0.000366

34980	Azaraphel	26	0.90625	0.948276	34.811905	0.000333
34988	SammySosa21 "sammysosa21"	25	0.878788	0.931034	22.657143	0.000318
34958	Mira Zarlax	17	0.707317	0.793103	3	0.000219
34959	Thomas Twain	17	0.707317	0.793103	3	0.000219
34976	Athos "Gamer Guy"	17	0.707317	0.793103	3	0.000219
34957	Doug E. White "Douglas White"	16	0.690476	0.775862	8.821429	0.000216
34977	Chris Rembisz	16	0.690476	0.775862	6.869048	0.000215
34979	mark twain	16	0.690476	0.775862	6.869048	0.000215

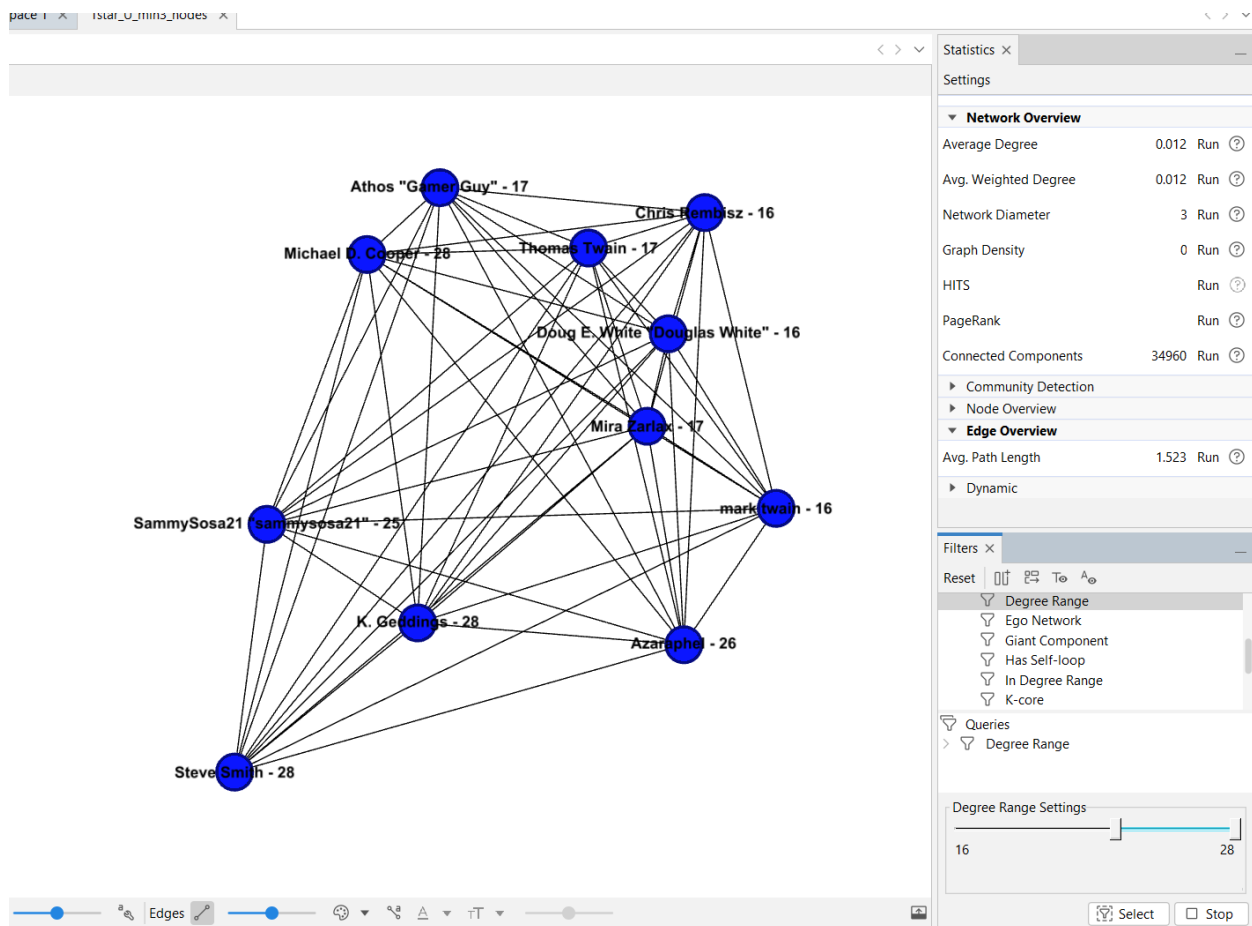


Figure 8 : Gephi 1 Star Set U min 3

Gephi – Set V

Table 18 : Gephi 1 star Set V min 3

Id	product title	Degree	Closeness centrality	Harmonic Closeness centrality	Betweenness centrality	Page ranks
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9658	Star Wars Galaxies: An Empire Divided	7	0.619048	0.74359	47.25	0.001252
9646	Spore	5	0.619048	0.692308	55	0.001392
9656	Star Wars Galaxies: Jump to Lightspeed Expansion Pack	5	0.448276	0.602564	0.25	0.000814
9667	Star Wars Galaxies: The Total Experience	5	0.448276	0.602564	0.25	0.000814
9672	Star Wars Galaxies: Starter Kit	5	0.448276	0.602564	0.25	0.000814
9660	Star Wars Galaxies 60 Day Pre-Paid Game Card	4	0.433333	0.564103	0	0.000668
9668	Star Wars Galaxies: The Complete Online Adventures	4	0.433333	0.564103	0	0.000668

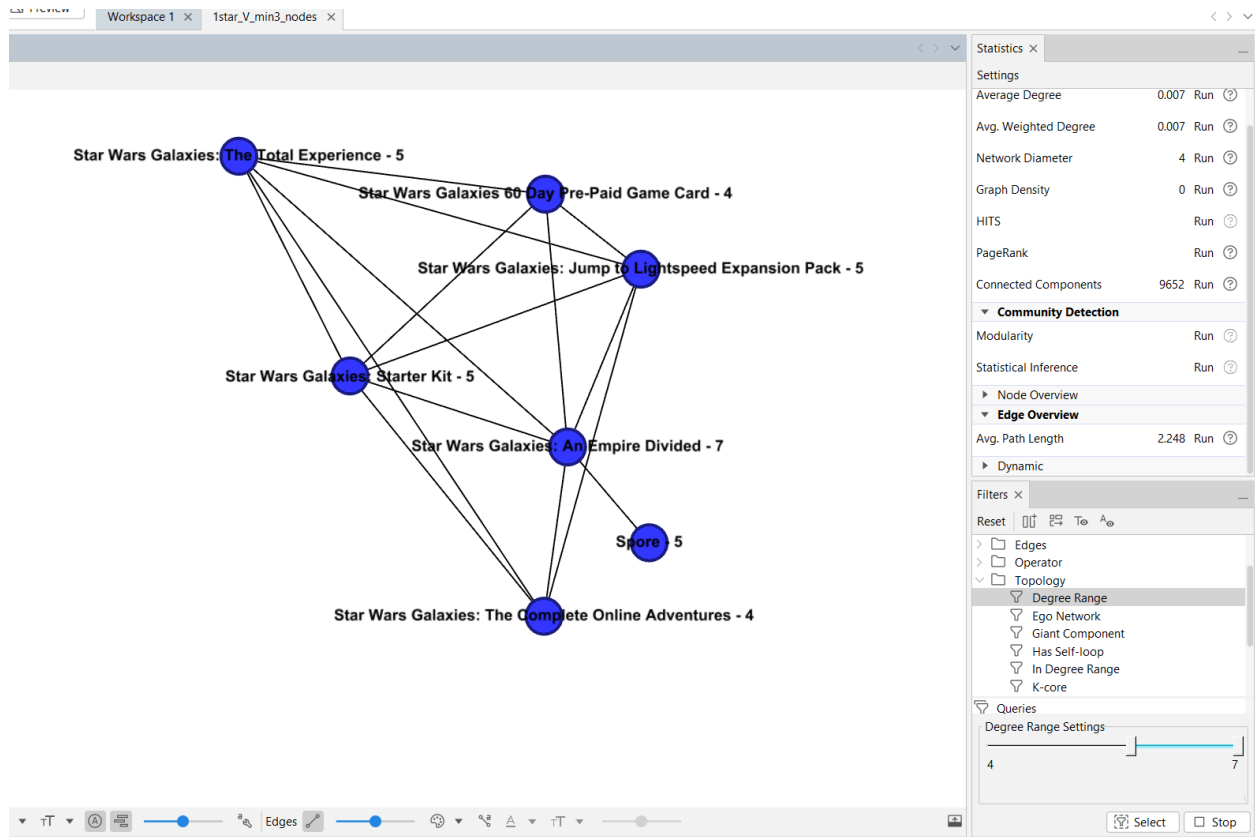


Figure 9 : Gephi 1 Star Set V min 3

Threshold Condition - Minimum 5 neighbours

5 Star

Gephi – Set U

Table 19 : Gephi 5 star Set U min 5

Id	profile name	Degree	Closeness centrality	harmonic closeness centrality	Betweenness centrality	Page ranks
130717	Lisa Shea	190	0.663516	0.762583	38960.53674	0.002217
130573	Cloud "..."	75	0.530211	0.590218	8349.533829	0.000669
130516	N. Durham "Big Evil"	64	0.516937	0.5717	6129.430484	0.000557
130545	Jack-of-All Games	62	0.516937	0.569801	7599.820147	0.000528
73307	S. Rhodes	54	0.513909	0.560304	3474.537368	0.000428
73283	Adam Sims	47	0.500713	0.545109	3692.851281	0.000359
73299	J. Parmelee	39	0.490223	0.530627	4119.577049	0.000323
73268	Samario C. Oliver	38	0.484807	0.525641	1725.655112	0.000315

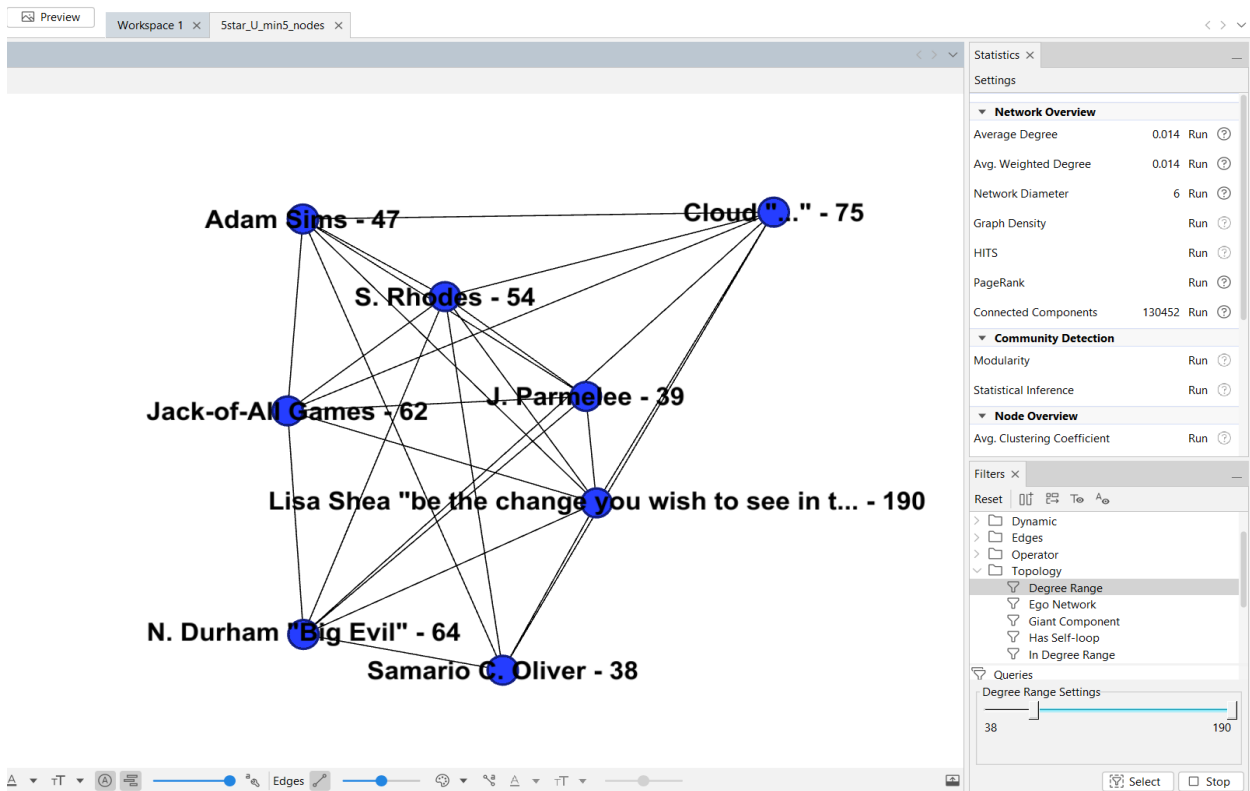


Figure 10 : Gephi 5 Star Set U min 5

Table 20 : Gephi 5 star Set V min 5

Id	product title	Degree	Closeness centrality	harmonic closeness centrality	Betweenness centrality	Page ranks
10247	Halo: Combat Evolved	92	0.452474	0.522309	35096.1806	0.004766
2126	Final Fantasy VII	84	0.451057	0.517448	22074.94652	0.003347
10276	GREATEST HITS:Grand Theft Auto III	79	0.449649	0.512905	24832.19596	0.00344
8246	Super Smash Bros Melee	78	0.44376	0.508767	14583.96033	0.003066
6197	Grand Theft Auto Vice City	76	0.444102	0.506597	15887.01202	0.003305
6170	Kingdom Hearts	60	0.427935	0.485041	10161.09323	0.002365
66	Wii	53	0.409673	0.467188	12760.82725	0.002295
8249	Grand Theft Auto: San Andreas	52	0.420438	0.473032	11474.989	0.002246
10271	Gamecube Console Platinum	51	0.426351	0.477951	6547.315175	0.001968

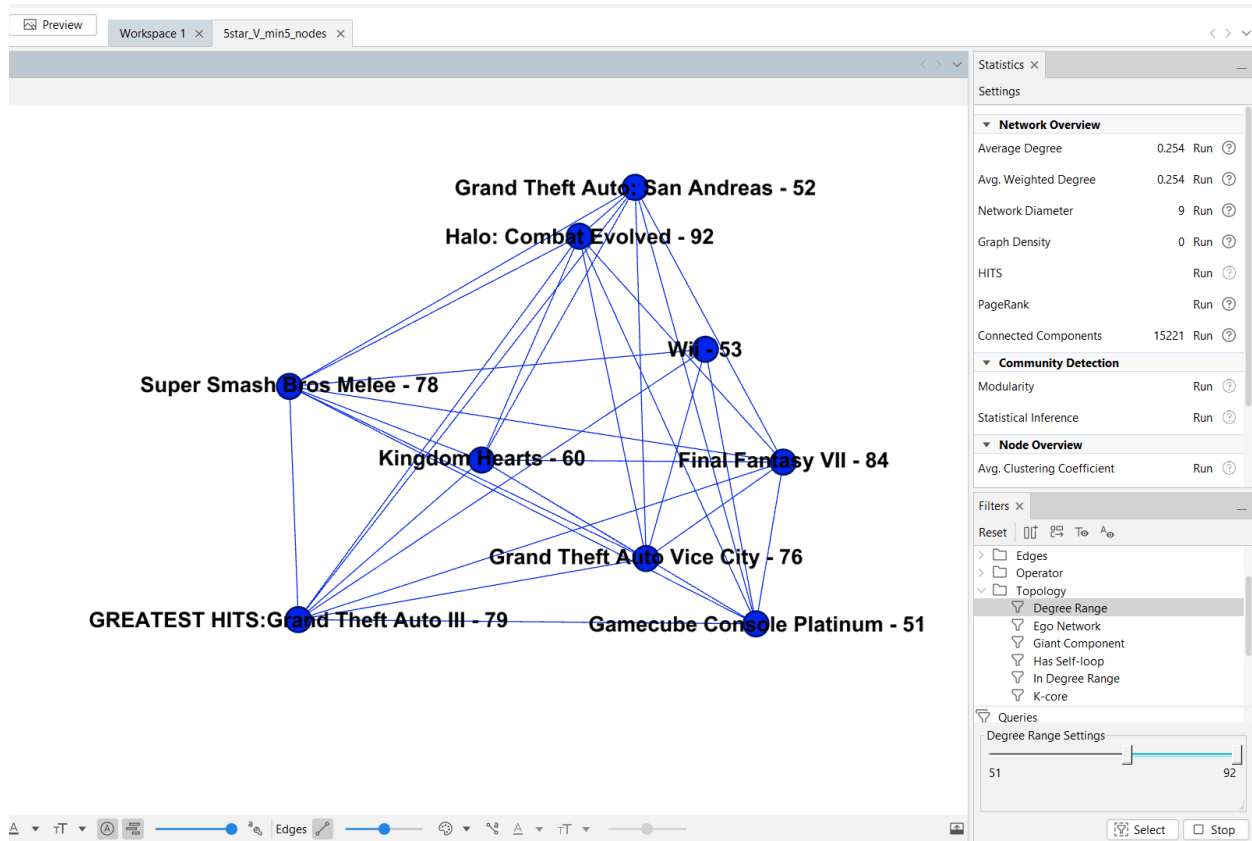


Figure 11 : Gephi 5 Star Set V min 5

4 Star

Gephi – Set U

Table 21 : Gephi 4 star Set U min 5

Id	Profile Name	Degree	Closeness centrality	harmonic closeness centrality	Betweenness centrality	Page ranks
58273	Lisa Shea	15	0.530303	0.674762	323.3333	0.000539
58251	Michael Kerner "Michael Kerner"	9	0.421687	0.552857	125.4167	0.000326
58275	S. Rhodes	8	0.460526	0.560476	115.6667	0.000276
58255	Cloud "..."	7	0.454545	0.535714	230.25	0.000262
58256	N. Durham "Big Evil"	7	0.448718	0.541429	61	0.000243
58259	Trent Foreman	5	0.397727	0.493333	52.25	0.000193
58270	SleepyJD	5	0.384615	0.479048	99	0.00022

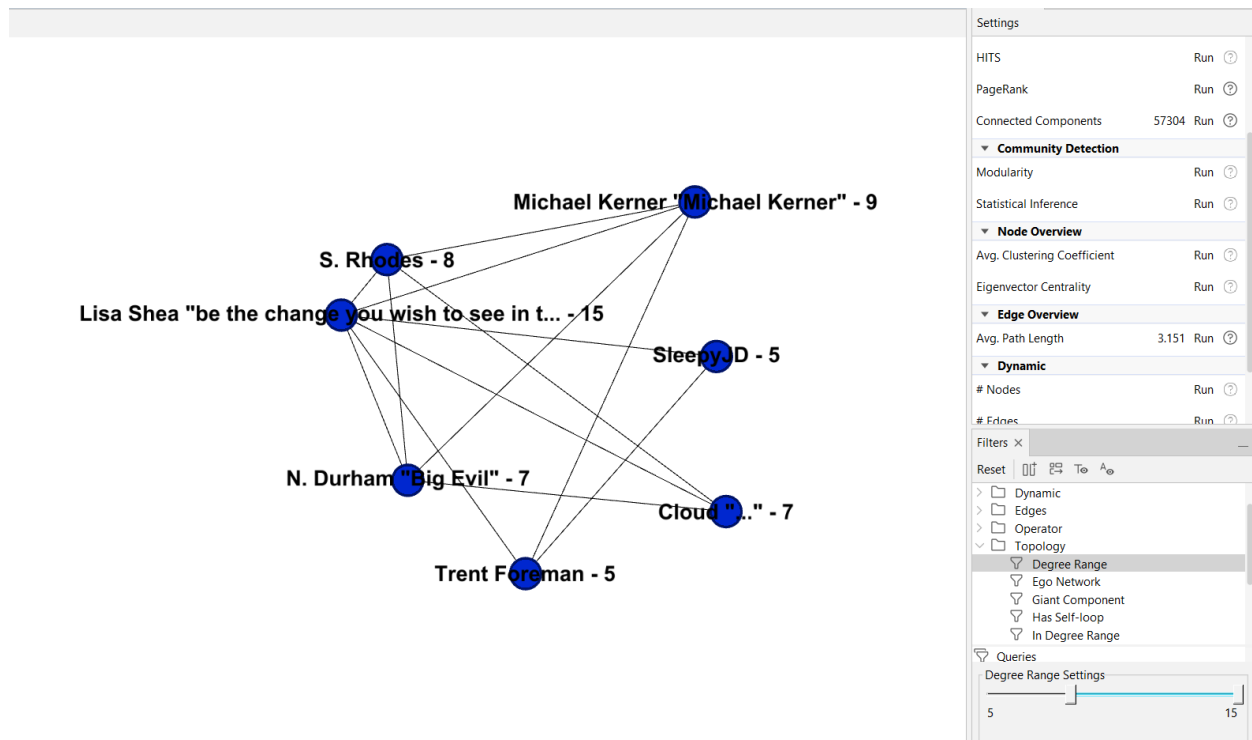


Figure 12 : Gephi 4 Star Set U min 5

Gephi – Set V

Table 22 : Gephi 4 star Set V min 5

Id	product title	Degree	Closeness centrality	harmonic closeness centrality	Betweenness centrality	Page ranks
13148	Final Fantasy X	5	0.777778	0.857143	15	0.000928
13128	Super Mario Sunshine	4	0.75	0.833333	13	0.001064
13107	Kingdom Hearts	3	0.583333	0.690476	6	0.000614
13140	Final Fantasy IX	3	0.583333	0.690476	2.5	0.000551
13151	Halo 3	3	0.714286	0.8	8	0.000828
13159	Final Fantasy VIII	3	0.583333	0.690476	2.5	0.000551

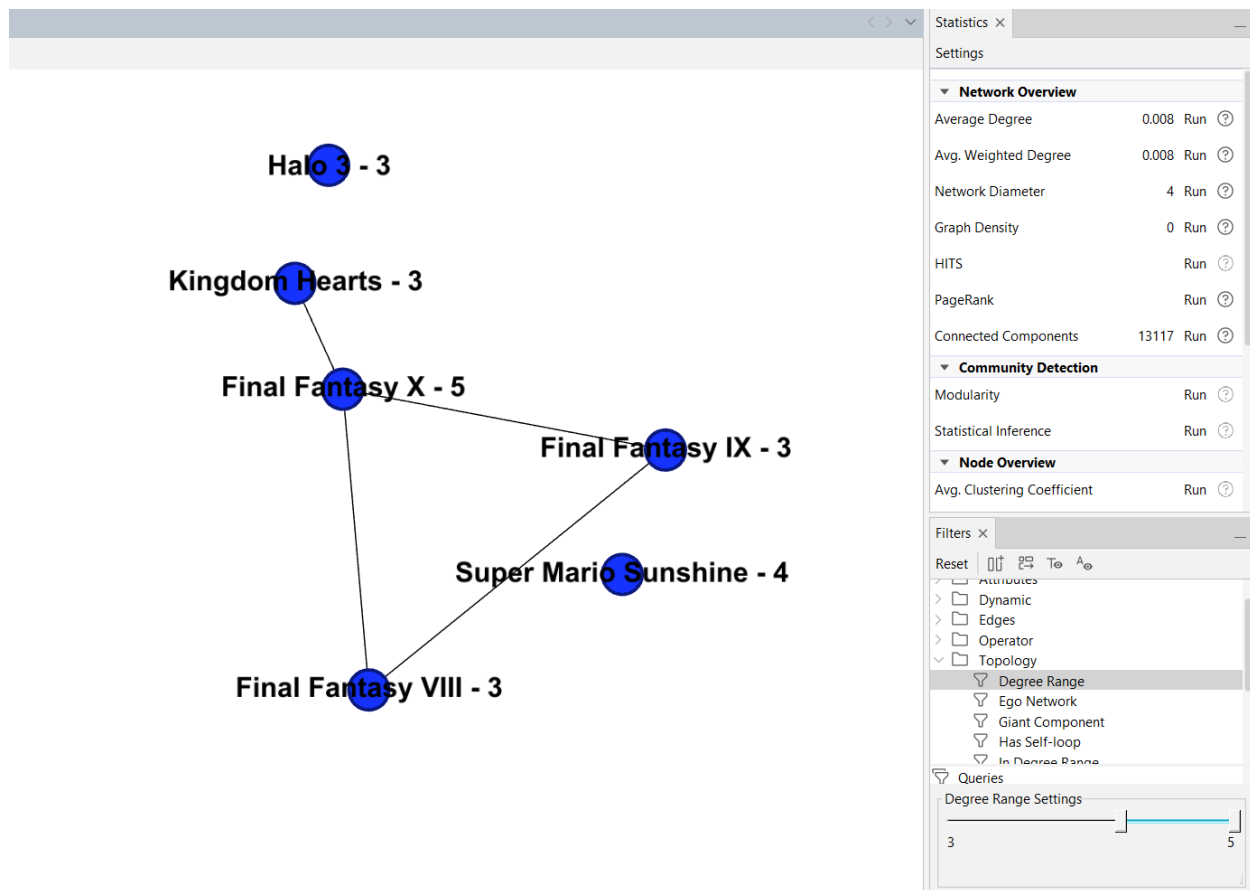


Figure 13 : Gephi 4 Star Set V min 5

3 Star

Gephi – Set U

Table 23 : Gephi 3 star Set U min 5

Id	Profile Name	Degree	Closeness centrality	harmonic closeness centrality	Betweenness centrality	Page ranks
28077	N. Durham "Big Evil"	2	1	1	1	0.000379
28079	S. Rhodes	1	0.666667	0.75	0	0.0002
28078	Lisa Shea	1	0.666667	0.75	0	0.0002

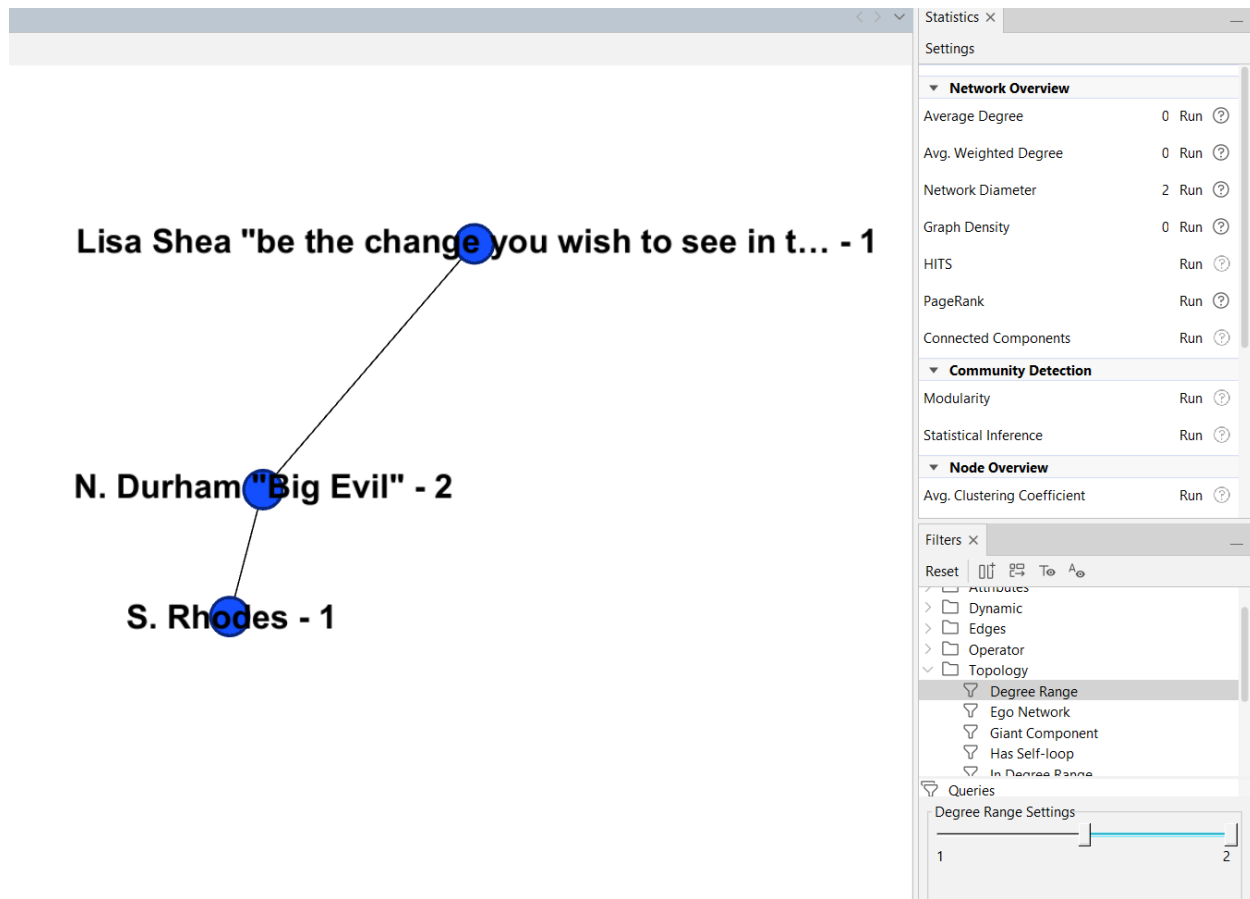


Figure 14 : Gephi 3 Star Set U min 5

Gephi – Set V

Table 24 : Gephi 3 star Set V min 5

Id	product title	Degree	Closeness centrality	harmonic closeness centrality	Betweenness centrality	Page ranks
9964	Monster Cable PLW100 FO-10 Fiber Optic Audio Hookup for Sony PlayStation 2	1	1	1	0	0.000664
9965	Xbox LIVE 12 Month Gold Membership Card	1	1	1	0	0.000664
9966	Monster Cable PLW100 FO-10 Fiber Optic Audio Hookup for Sony PlayStation 2	1	1	1	0	0.000664
9967	Xbox LIVE 3 Month Gold Membership	1	1	1	0	0.000664

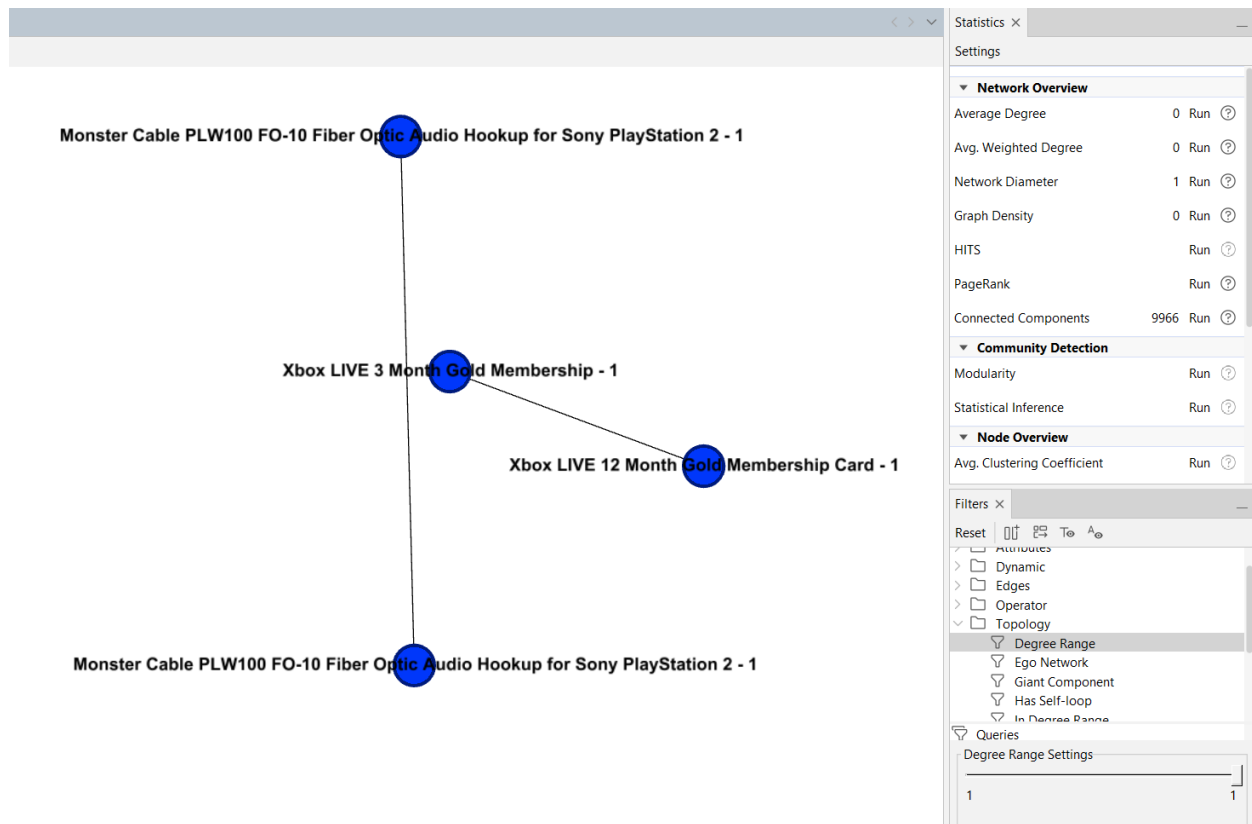


Figure 15 : Gephi 3 Star Set $V_{min 5}$

2 Star
Gephi Set U – no edges
Gephi Set V

Table 25 : Gephi 2 star Set $V_{min 5}$

Id	product title	Degree	Closeness centrality	Harmonic Closeness Centrality	Betweenness centrality	Page ranks
7552	NBA Live 2001	1	1	1	0	0.000873
7553	Xbox LIVE 3 Month Gold Membership	1	1	1	0	0.000873
7554	Monster Cable PLW100 FO-10 Fiber Optic Audio Hookup for Sony PlayStation 2	1	1	1	0	0.000873
7555	NBA Live 2001	1	1	1	0	0.000873
7556	Xbox LIVE 12 Month Gold Membership Card	1	1	1	0	0.000873
7557	Monster Cable PLW100 FO-10 Fiber Optic Audio Hookup for Sony PlayStation 2	1	1	1	0	0.000873

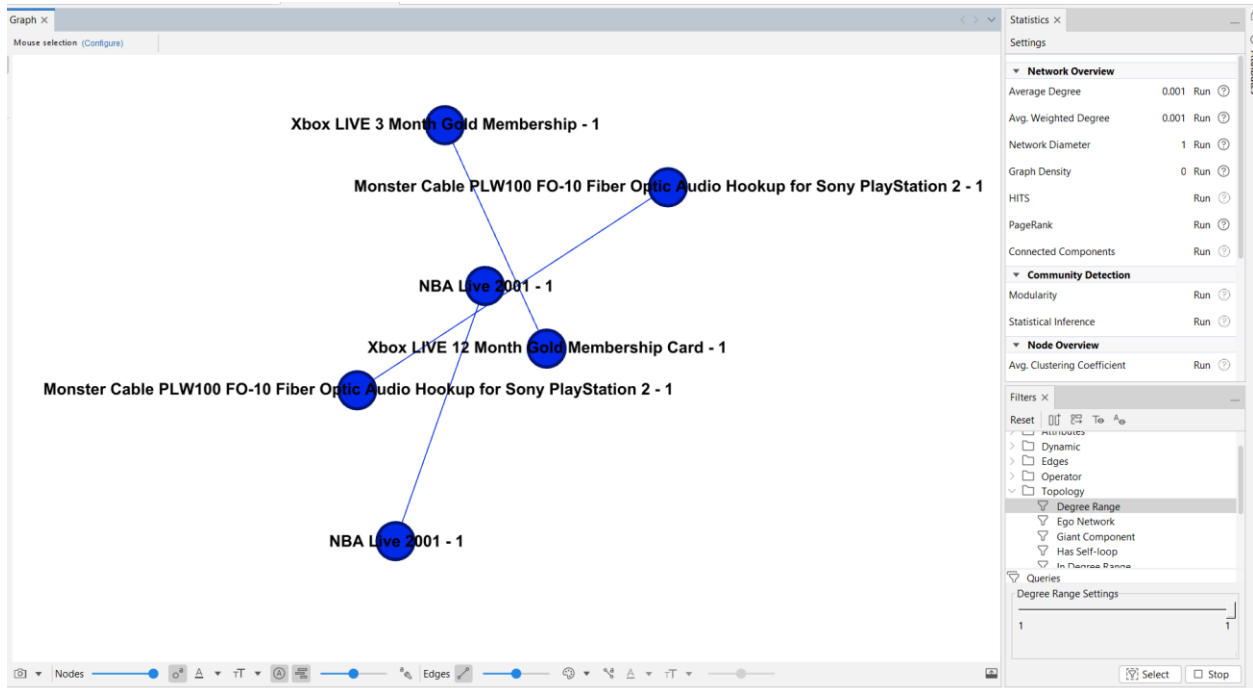


Figure 16 : Gephi 2 Star Set V min 5

1 Star

Gephi – Set U

Table 26 : Gephi 1 star Set U min 5

Id	Profile Name	Degree	Closeness centrality	harmonic closeness centrality	Betweenness centrality	Page ranks
34989	K. Geddings	2	1	1	0	0.000189
34991	Michael D. Cooper	2	1	1	0	0.000189
34993	Steve Smith	2	1	1	0	0.000189
34990	A.Crichton	1	1	1	0	0.000189
34992	Josh X	1	1	1	0	0.000189

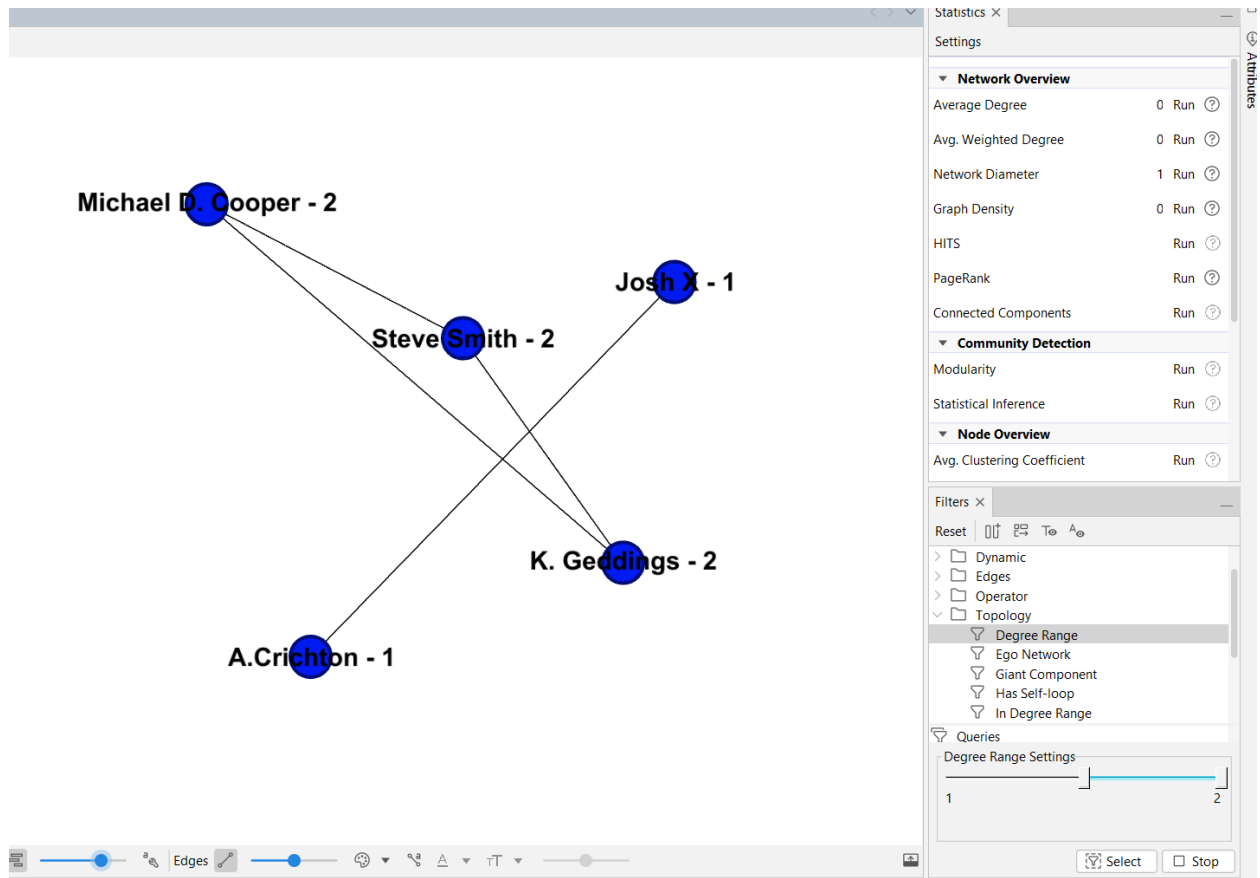


Figure 17 : Gephi 1 Star Set U min 5

Gephi – Set V

Table 27 : Gephi 1 star Set V min 5

Id	product title	Degree	Closeness centrality	harmonic closeness centrality	Betweenness centrality	Page ranks
9661	Star Wars Galaxies: Jump to Lightspeed Expansion Pack	5	1	1	0.25	0.000722
9663	Star Wars Galaxies: An Empire Divided	5	1	1	0.25	0.000722
9669	Star Wars Galaxies: The Total Experience	5	1	1	0.25	0.000722
9673	Star Wars Galaxies: Starter Kit	5	1	1	0.25	0.000722
9665	Star Wars Galaxies 60 Day Pre-Paid Game Card	4	0.833333	0.9	0	0.000593
9670	Star Wars Galaxies: The Complete Online Adventures	4	0.833333	0.9	0	0.000593

9657	Spore	2	1	1	1	0.000991
9668	Xbox 360 Pro Console 20GB [Old Version]	2	1	1	1	0.000991

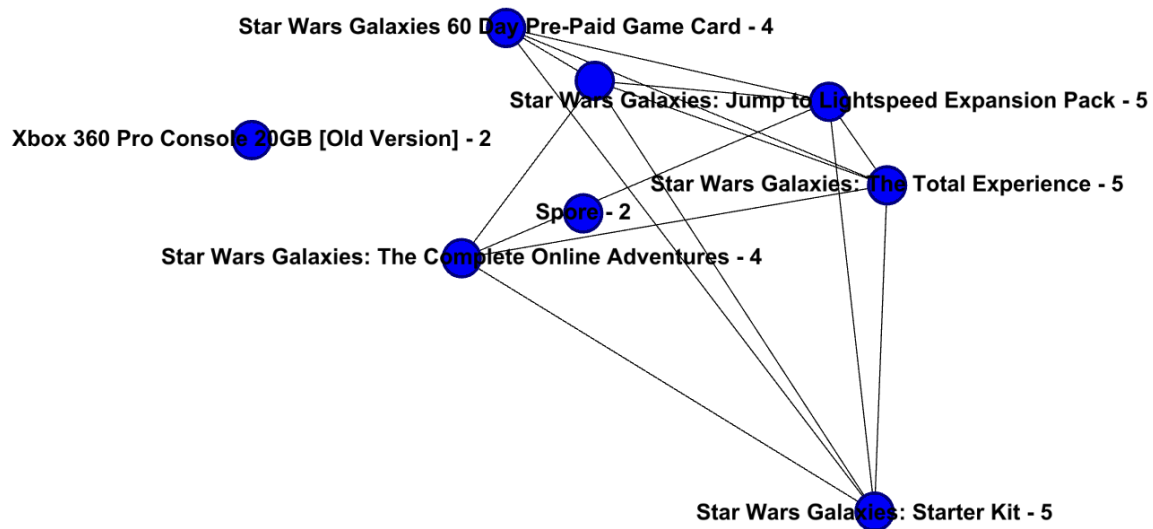


Figure 18 : Gephi 1 Star Set V min 5

5. DISCUSSION

Ratings and Network Connectivity

The results show that the review networks become richer as the star rating goes up. The 1 and 2-star networks are almost empty, the 3-star network is still quite sparse, the 4-star network starts to show real clustering, and the 5-star network is by far the most connected. A likely reason is the way positive and negative reviewers behave differently. People leaving negative reviews rarely have games in common, for example if someone dislikes a game, they usually just leave a review and move on rather than seeking out others who felt the same. Positive reviewers, on the other hand, cluster strongly around the popular titles, which creates many shared connections. This is also why the 1 and 2-star networks almost disappear once the min5 threshold is applied. The product (game) networks were also denser than the user networks, meaning the top-rated games group together more tightly than the people reviewing them.

The Role of Hubs

Within the most connected networks, especially the 5-star network, a small number of nodes stand out as hubs. These hubs are mostly the games themselves. Popular titles such as GTA III and Halo: Combat Evolved have the highest degree, PageRank, and betweenness in the product network. They act as bridges, connecting otherwise separate groups of players who would not share any games in common. This matches the idea of scale-free networks described by Albert and Barabási [8], where a few nodes end up with far more connections than the rest. In practice these hubs are useful. They point to the games that hold a community together, which can help with marketing and with understanding how players cluster around top-rated titles.

Effects of the Threshold (Min 3 and Min 5)

Every network was built in two versions. In the first, two nodes connect only if they share at least three

neighbours, and in the second they need at least five. This stricter rule makes a big difference. For the 5-star network, moving from three to five shared neighbours dropped the number of edges from over 15,000 down to 888. The connections that survive are the strongest ones, meaning the pairs of nodes that share the most in common. Because the weaker links are removed, the network shrinks to a small, tightly connected core, which is why the diameter drops. Overall, this shows that how connected a node looks depends a lot on how strict the rule is for what counts as a connection.

6. CONCLUSION

This project analyzed how Amazon video game products from 1995 to 2013 were reviewed by customers, using a bipartite network of customers and games. The reviews were grouped by star rating, and for each rating the user and product projections were compared at two thresholds, min3 and min5, using measures such as density, clustering, diameter, and community structure.

The main finding is that higher star ratings produce richer and more connected networks, while lower ratings stay sparse and fragmented. The 1 and 2-star networks were almost empty, and the networks became steadily more connected up to the 5-star rating. Within these networks, a small number of popular games acted as hubs that bridged different communities of players. Comparing the two thresholds showed that the structure and the hubs depend heavily on how a connection is defined: the stricter min5 rule removed the weaker links and left a smaller, tightly connected core.

There are several directions for future work. The same analysis could be applied to a different time period to see whether these patterns hold or extended to include customer and genre preferences for a deeper picture. The approach could also be applied to completely different datasets. A few refinements would improve the analysis further, such as using unique user IDs instead of profile names so that different people are not treated as one, adding sentiment analysis to capture what reviewers say, and using the product networks to suggest related games.

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Appendix

```
1 #Set U projection (min 3 and min 5):
2 CREATE OR REPLACE TABLE edges_base AS
3 SELECT DISTINCT
4     "User ID" AS u_col,
5     "Product ID" AS v_col
6 FROM raw_data
7 WHERE "User ID" IS NOT NULL
8     AND "Product ID" IS NOT NULL;
9
10 COPY (
11     SELECT e1.u_col AS Source, e2.u_col AS Target, COUNT(*) AS Shared_Products
12     FROM edges_base e1
13     JOIN edges_base e2 ON e1.v_col = e2.v_col
14     WHERE e1.u_col < e2.u_col
15     GROUP BY e1.u_col, e2.u_col
16     HAVING COUNT(*) >= {MIN_SHARED_NEIGHBORS}
17 ) TO 'Set_U_Users_Edges.csv' (HEADER, DELIMITER ',');
18 #Set V projection (min 3 and min 5):
19 COPY (
20     SELECT e1.v_col AS Source, e2.v_col AS Target, COUNT(*) AS Shared_Users
21     FROM edges_base e1
22     JOIN edges_base e2 ON e1.u_col = e2.u_col
23     WHERE e1.v_col < e2.v_col
24     GROUP BY e1.v_col, e2.v_col
25     HAVING COUNT(*) >= {MIN_SHARED_NEIGHBORS}
26 ) TO 'Set_V_Products_Edges.csv' (HEADER, DELIMITER ',');
27 #Creation of Projection sets
28 # Set U Edges & Nodes:
29 print("\n[1/2] Projecting User Network (Set U)...")
30
31 con.execute(f"""
32 COPY (
33     SELECT e1.u_col AS Source, e2.u_col AS Target, COUNT(*) AS Shared_Products
34     FROM edges_base e1
35     JOIN edges_base e2 ON e1.v_col = e2.v_col
36     WHERE e1.u_col < e2.u_col
37     GROUP BY e1.u_col, e2.u_col
38     HAVING COUNT(*) >= {MIN_SHARED_NEIGHBORS}
39 ) TO 'Set_U_Users_Edges.csv' (HEADER, DELIMITER ',');
40 """)
41
42 print("Generating User Node Table...")
43 con.execute("""
44 CREATE TEMP TABLE proj_u_edges AS SELECT Source, Target FROM read_csv_auto('Set_U_Users_Edges.csv');
45 CREATE TEMP TABLE deg_u AS
46 SELECT Node, COUNT(*) AS Projected_Degree FROM (
47     SELECT Source AS Node FROM proj_u_edges
48     UNION ALL
49     SELECT Target AS Node FROM proj_u_edges
50 ) GROUP BY Node;
51 COPY (
52     SELECT
53         u.Node_ID AS "User ID",
54         u.Profile_Name AS "Profile Name",
55         COALESCE(d.Projected_Degree, 0) AS Projected_Degree
56     FROM user_attributes u
57     LEFT JOIN deg_u d ON u.Node_ID = d.Node
58 ) TO 'Set_U_Users_Nodes.csv' (HEADER, DELIMITER ',');
59 DROP TABLE proj_u_edges; DROP TABLE deg_u;
60 """)
```

```

1  # Set V Edges & Nodes:
2  print("\n[2/2] Projecting Product Network (Set V)...")
3
4  con.execute(f"""
5  COPY (
6      SELECT e1.v_col AS Source, e2.v_col AS Target, COUNT(*) AS Shared_Users
7      FROM edges_base e1
8      JOIN edges_base e2 ON e1.u_col = e2.u_col
9      WHERE e1.v_col < e2.v_col
10     GROUP BY e1.v_col, e2.v_col
11     HAVING COUNT(*) >= {MIN_SHARED_NEIGHBORS}
12 ) TO 'Set_V_Products_Edges.csv' (HEADER, DELIMITER ',');
13 """)
14
15 print("Generating Product Node Table...")
16 con.execute("""
17 CREATE TEMP TABLE proj_v_edges AS SELECT Source, Target FROM read_csv_auto('Set_V_Products_Edges.csv');
18 CREATE TEMP TABLE deg_v AS
19 SELECT Node, COUNT(*) AS Projected_Degree FROM (
20     SELECT Source AS Node FROM proj_v_edges
21     UNION ALL
22     SELECT Target AS Node FROM proj_v_edges
23 ) GROUP BY Node;
24 COPY (
25     SELECT
26         p.Node_ID AS "Product ID",
27         p.Product_Title AS "Product Title",
28         COALESCE(d.Projected_Degree, 0) AS Projected_Degree
29     FROM product_attributes p
30     LEFT JOIN deg_v d ON p.Node_ID = d.Node
31 ) TO 'Set_V_Products_Nodes.csv' (HEADER, DELIMITER ',');
32 DROP TABLE proj_v_edges; DROP TABLE deg_v;
33 """)
34 #Metric Computations
35 #Density, clustering, global efficiency, LCC path/diameter
36 density = nx.density(G) if n > 1 else 0.0
37 avg_clustering = nx.average_clustering(G) if n > 0 else 0.0
38 try:
39     global_eff = nx.global_efficiency(G) if n > 1 else 0.0
40 except:
41     global_eff = None
42 # LCC-only metrics
43 lcc_n = G_lcc.number_of_nodes()
44 lcc_m = G_lcc.number_of_edges()
45 lcc_avg_shortest_path = None
46 lcc_diameter = None
47 lcc_avg_clustering = None
48 lcc_transitivity = None
49
50 if lcc_n > 1 and lcc_m > 0:
51     try:
52         lcc_avg_shortest_path = nx.average_shortest_path_length(G_lcc)
53     except:
54         lcc_avg_shortest_path = None
55     try:
56         lcc_diameter = nx.diameter(G_lcc)
57     except:
58         lcc_diameter = None

```

```

1 #Nodes, edges, top-degree, top-harmonic, top-closeness, top-betweenness
2 n = G.number_of_nodes()
3 m = G.number_of_edges()
4 isolates = set(nx.isolates(G))
5 components = list(nx.connected_components(G)) if n > 0 else []
6 largest_cc_nodes = max(components, key=len) if components else set()
7 G_lcc = G.subgraph(largest_cc_nodes).copy() if largest_cc_nodes else nx.Graph()
8 degree_dict = dict(G.degree())
9 harmonic = nx.harmonic_centrality(G) if n > 0 else {}
10 def safe_closeness_lcc(G_lcc, exact_threshold=8000):
11     n = G_lcc.number_of_nodes()
12     if n == 0:
13         return {}, "skipped_empty"
14     try:
15         if n <= exact_threshold:
16             closeness = nx.closeness_centrality(G_lcc)
17             return closeness, "exact_lcc"
18         else:
19             return {}, "skipped_too_large"
20     except:
21         return {}, "error"
22
23 def safe_betweenness_lcc(G_lcc, exact_threshold=3000, approx_threshold=15000, k_sample=500):
24     n = G_lcc.number_of_nodes()
25     if n == 0:
26         return {}, "skipped_empty"
27     try:
28         if n <= exact_threshold:
29             betweenness = nx.betweenness_centrality(G_lcc, normalized=True)
30             return betweenness, "exact_lcc"
31         elif n <= approx_threshold:
32             k = min(k_sample, n)
33             betweenness = nx.betweenness_centrality(G_lcc, k=k, normalized=True, seed=42)
34             return betweenness, f"approx_lcc_k{k}"
35         else:
36             return {}, "skipped_too_large"
37     except:
38         return {}, "error"
39 # LCC centralities
40 closeness_lcc, closeness_method = safe_closeness_lcc(G_lcc, exact_threshold=8000)
41 betweenness_lcc, betweenness_method = safe_betweenness_lcc(
42     G_lcc,
43     exact_threshold=3000,
44     approx_threshold=15000,
45     k_sample=500
46 )
47 # Node-level results
48 node_rows = []
49 for node in G.nodes():
50     node_rows.append({
51         ...
52         "closeness_lcc_only": closeness_lcc.get(node, None) if node in largest_cc_nodes else None,
53         "betweenness_lcc_only": betweenness_lcc.get(node, None) if node in largest_cc_nodes else None,
54     })
55 node_metrics_df = pd.DataFrame(node_rows)

```

```

1  #Helper for top nodes
2  def top_node(df, col):
3      df2 = df.dropna(subset=[col]).sort_values(col, ascending=False)
4      if len(df2) == 0:
5          return (None, None, None)
6      r = df2.iloc[0]
7      return (r["node_id"], r["node_label"], r[col])
8  #Top node values
9  top_degree_id, top_degree_label, top_degree_val = top_node(node_metrics_df, "degree")
10 top_harm_id, top_harm_label, top_harm_val = top_node(node_metrics_df, "harmonic_centrality")
11 top_close_id, top_close_label, top_close_val = top_node(node_metrics_df, "closeness_lcc_only")
12 top_betw_id, top_betw_label, top_betw_val = top_node(node_metrics_df, "betweenness_lcc_only")
13 # Result validation checks
14 def validate_graph_results(G, graph_name, summary_dict, components, nodes_df, edges_df,
15                             node_id_col, source_col, target_col):
16     n = G.number_of_nodes()
17     m = G.number_of_edges()
18     degree_sum = sum(dict(G.degree()).values())
19     manual_density = (2 * m) / (n * (n - 1)) if n > 1 else 0.0
20     ...
21     checks = {
22         "graph_name": graph_name,
23         "check_num_nodes_matches_graph_object": (summary_dict["num_nodes"] == n),
24         "check_num_edges_matches_graph_object": (summary_dict["num_edges"] == m),
25
26         # degree sum = 2 * number of edges
27         "check_degree_sum_equals_2m": (degree_sum == 2 * m),
28
29         # Sum of component sizes = total nodes
30         "check_component_sizes_sum_to_n": (sum(len(c) for c in components) == n),
31
32         # average clustering and global efficiency in [0,1]
33         "check_average_clustering_in_range": (
34             summary_dict["average_clustering"] is None or
35             0 <= summary_dict["average_clustering"] <= 1
36         ),
37         "check_global_efficiency_in_range": (
38             summary_dict["global_efficiency"] is None or
39             0 <= summary_dict["global_efficiency"] <= 1
40         ),
41     }
42

```